

# Benchmark for Crack Re-Identification in Ticket Management Systems

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**Abstract**—In a ticket-based maintenance system, each new photograph of a crack should be linked to the correct ticket. However, the same crack photographed from a different angle may be mistaken for a new defect, breaking its history and leading to unnecessary inspection work. We introduce CRACKID, a dataset designed to study this problem of identifying the same physical crack across photographs. The dataset contains 140 photographs from 17 walls, covering 212 distinct crack identities labelled using homography-based annotation. We evaluate ten off-the-shelf image-matching methods across three tasks: finding the correct photograph, deciding whether two photographs show the same crack, and rejecting photographs of previously unseen cracks. Although some methods retrieve the correct crack reasonably well, their decisions about whether two images show the same crack are only slightly better than a simple constant prediction. Experiments with surrounding wall context suggest that much of the matching signal may come not from the crack itself, but from its surroundings, including nearby cracks. A geometric approach that aligns photographs using the surrounding wall surface performs substantially better on the images that can be aligned, highlighting the potential value of geometric information for reliable crack association. These results show that good retrieval performance alone does not guarantee reliable crack association. CRACKID provides a foundation for developing more dependable visual methods for connecting crack photographs across maintenance records, with cross-visit evaluation and ticket-level duplicate reduction as important next steps.

**Index Terms**—Annotation Quality, Benchmark, Building Maintenance, Crack Re-Identification, Geometric Alignment, Homography, Structural Inspection

## I. INTRODUCTION

Ticket management systems keep records of reports regarding physical defects so that these reports can be examined, given priority, repaired, and followed up on. In principle, each ticket should contain the full history of one particular incident. In reality, different users may take more than one photograph of the same crack, which results in duplicate tickets and breaks up the history of the incident and leads to unnecessary assessment work. A new photograph must therefore either be linked to an existing ticket or a new ticket have to be opened for any damage that has not yet been recorded.

This is a case of visual re-identification (re-ID). It is necessary to identify the same physical crack even though there are variations in the angle of view, the distance from the camera, the field of view, the lighting conditions, and partial visibility. Furthermore, a crack may change by extending, widening or branching, although such changes are not considered here: the protocol supposes that the morphology remains unaltered

during a single visit and only assesses associations. Genuine new damage must be differentiated from existing damage. Making errors in either direction results in duplicate tickets or in the incorrect assignment of a damage history. A crack identity consists of the set of photographs of a single physical crack. In practice, one crack identity corresponds to one ticket, but this benchmark focuses on visual association rather than on reducing duplicates at the ticket level.

Existing work on crack detection and segmentation focuses on single photographs. Datasets such as CrackSeg9k [1] and OmniCrack30k [2] contain no cross-photograph identities, so photographs cannot be associated with tickets. It is unclear whether person and object re-ID techniques [3], [4] transfer to cracks: different cracks can look alike, and one crack can change appearance considerably.

This study presents CRACKID as a method for determining whether photographs show the same physical crack: it includes 140 photographs taken of 17 walls during one visit. Since the photographs are real, they include the changes that occur during the visit. The accuracy figures given are point estimates based on a small sample, with the samples grouped by wall.

We use CRACKID to evaluate ten off-the-shelf matchers. The protocol divides the process into two parts: retrieving the gallery candidates and then deciding whether two photographs display the same crack. It involves open-set identification with rejection, meaning that the top match is only accepted if it exceeds a certain threshold, and in all other cases the query is treated as a new ticket. Having a high rank by itself is not sufficient to link a report to a ticket. In order to determine whether the matchers recognise the crack itself or merely the surrounding surface context, we employ crack-erased controls together with a geometric reference that aligns the wall photographs and compares the projected crack masks.

Three research questions guide the study:

**RQ1.** Can the evaluated methods associate photographs of the same crack taken under different acquisition conditions?

**RQ2.** How do wall context and geometric alignment contribute to re-identification within a single visit?

**RQ3.** How do two selected approaches perform under synthetic changes in appearance and crack growth?

RQ1 and RQ2 use real single-visit photographs; RQ3 is an exploratory synthetic probe.

The following points make up our contribution: (1) CRACKID: 140 photographs, 17 walls, 212 identities (37 of

which are matchable), along with a wall-disjoint protocol for retrieval and open-set identification with rejection (RQ1); (2) a registration-plus-Chamfer reference that maintains the alignment target (i.e. the wall texture excluding the cracks) separate from the identified target (i.e. the projected crack masks), together with context controls and coverage analysis (RQ2); (3) an evaluation of ten off-the-shelf matchers using controls that isolate crack appearance from the surrounding surface context; and (4) a synthetic revisit probe for RQ3. The accuracy figures are point estimates obtained from a small sample of wall-clustered data; the effective sample consists of the walls, not the queries or pairs.

## II. RELATED WORK

Most crack work detects/segments single photographs [5]–[7]. Consolidated sets [1], [2] give no cross-photograph identities. Multi-view pipelines fuse one inspection via geometry [8] without persistent identities. Monitoring tracks fixed cameras [9]–[11] or guides (patent [12]) with identities at construction. Ours combines object re-ID [4] with open-set identification [13]; ranking follows Market-1501 [14].

Standard blocks are SIFT [15], ORB [16], SuperGlue [17] and LoFTR [18] (indoor weights), crop embeddings [3], [19]–[22] and YOLOv8 [23]. Outdoor weights, LightGlue, dense matchers, full-photo descriptors (e.g. NetVLAD [24]) and matcher-fed Chamfer are not tested. Vehicle damage [25] shares the open-set protocol; other damage work is detection/severity [26]–[28]. Align-first is change detection [29], [30] via planar registration [31]–[33] with ECC [34]; spatial verification [35], [36] is the precedent. The contribution is protocol/labels/controls, not the machinery. Location/BIM/markers/SfM are unevaluated, though the wall is given. No public crack re-ID benchmark found in our searches. Alignment (wall) and matching (crack) targets must be disjoint.

## III. DATA ACQUISITION

### A. Acquisition and composition

The CRACKID pilot dataset consists of 140 photographs of 17 walls taken at a single site over four half-days with two consumer handheld devices, archived at Mendeley Data [37] (frozen snapshot; commit hash in Sec. IV); see Fig. 1. Device, resolution and lighting metadata are not archived per photograph. Walls are stippled render (textured) or smooth plaster (feature-poor); no surface roster is archived, so surface is proxied by registration rate and sharpness (variance of Laplacian at quarter scale): walls with median sharpness  $\leq 17$  register at 0.00–0.11, walls with  $\geq 62$  at 0.29–1.00. Six walls serve for validation (01–06) and eleven for testing (07–17). The test split has 74 photographs forming 301 same-wall pairs, of which 97 register (rate 0.32; 0.00 on walls 10 and 17 to 1.00 on walls 07, 08, 12). Retrieval uses known queries (19 test identities; 280 queries); open set adds queries with no gallery mate (258 on test). Walls 18–19 are excluded; the live labels directory has since diverged and must not be combined with frozen figures without re-running. Code is at

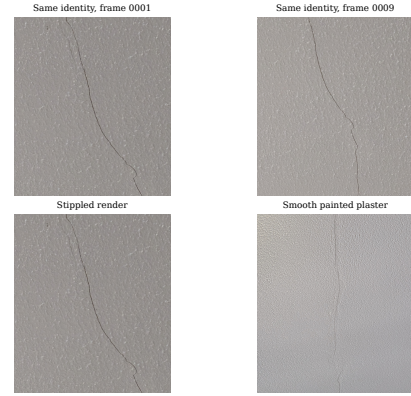


Fig. 1. CRACKID examples. Top: one identity across non-adjacent frames. Bottom: stippled vs smooth plaster. Homography-propagated labels, single annotator, no audit.

<https://github.com/DelcioKelson/re-id> (no code licence stated). Each wall was photographed in a continuous walk-around (median 3 s between frames, max 81 s span), an easier setting than multi-visit matching. Cross-wall search is disallowed, so the wall is given for free.

### B. Annotation and its limits

The annotation process begins with a labelling tool that applies Oriented FAST and Rotated BRIEF (ORB) [16] homography to propagate each click across the different frames and then combines the resulting fragments into crack identities. There is no gap threshold preserved and no written rule for merging; instead, one annotator decided to combine the propagated fragments that were judged to be part of the same physical crack. Merging represents the greatest risk since a wrong merge can cause a clear negative to be changed into a false positive. In the frozen snapshot, 95 out of the 212 identities (45%; 70 out of 146, 48% on the test set) are the result of merges, the median bridged gaps being 286 px (in the validation set) and 228 px (in the test set) where recorded. That annotator looked at all the clicks and categorised the unclaimed points (107 out of 1,023 were classified as no crack); agreement was not assessed and no second annotator carried out labelling without the homography overlays. Since the geometric reference is also aligned by means of homography, its benefit might be in measuring the agreement between two homography pipelines and one annotator’s merges rather than in determining crack identity (see Sec. VIII). All the figures refer to the frozen Mendeley snapshot. The positive pool in the frozen state is almost degenerate in that 5,718 of the 5,722 positive pairs involve a merged identity. A subset of the data that has been labelled independently by a second annotator is required in order to quantify the error due to merging.

## IV. BENCHMARK PROTOCOL

The benchmark tests visual association: for a new crack photograph, decide whether it shows a recorded crack. Annotated crack identities stand in for ticket records. Scores there-

fore measure association accuracy, not ticket-level duplicate reduction.

#### A. Task formulation and candidate records

A single identity is established by using photographs of the same physical crack. By 'instance' is meant one mask component in a single photograph; the 'query' is the instance that is being linked; a 'valid pair' is a query and a gallery cell that is accepted by the mask (specifically, the same wall but taken at a different time); a 'photograph block' consists of all the cells from one photograph pair and is the unit used for Hungarian assignment. Search across walls is not permitted. A query is considered 'known' if its gallery includes its identity; otherwise it is 'unknown' (in this case it is a singleton within one photograph only; singletons can differ in size or border distance, so FAR might indicate the rejection of fragments; see Sec. VIII). In closed-set retrieval only ranking is performed; in open-set identification links are made to the best ticket or the query is rejected as a new ticket. Pairs are the unit for making decisions; queries are used in retrieval and in open-set identification; photograph pairs are used for registration and assignment. The Hungarian one-to-one matching is applied separately to each photograph pair; the assignment  $F_1$  is calculated but is not presented as a headline result.

Counts for the frozen snapshot (6 validation walls 01–06; 11 test walls 07–17; walls 18–19 excluded): validation has 6 walls / 66 photographs / 66 identities (18 matchable, 439 known queries, 74 open-set); test has 11 walls / 74 photographs / 146 identities (19 matchable, 280 known queries, 258 open-set); positive query–gallery pairs: 26,012 validation / 5,722 test; valid query–gallery pairs: 89,118 / 31,822. Of 301 same-wall test photograph pairs, 255 have valid cells and 97 register successfully. Of 31,822 valid test query–gallery pairs, 15,328 fall inside registered pairs (48%).

#### B. Test conditions, decisions, and metrics

The main test involves actual walk-arounds assuming that the morphology remains unaltered; the variables included are viewpoint, distance, scale, lighting and visibility, as well as same-wall distractors and unknowns, with no separation being made according to gap, scale or surface. The validation and test walls are not connected. The appearance methods are run using the version released on the shared masks without any CRACKID tuning; the geometry settings (tolerance [2, 16] px, early-rejection distance, sharpness gate, Random Sample Consensus (RANSAC) [32] inliers, scale [1/16, 16]) are not fixed during validation and should be set before any final comparison. The masks are derived from a UNet++ [38], [39] combined with a ResNet-34 [40] that was trained on cracks available from the public dataset only (without use of a CRACKID wall), and this is taken as given with the error left unmeasured. For ORB/SIFT the RANSAC inliers are used, for SuperGlue/LoFTR the indoor match counts are employed, and for the embeddings the cosine value on L2 crops is taken. The inpainting control makes use of Telea after a 5 px dilation (see Sec. VI). The chance value is  $2p/(1+p)$

together with 200 random matrices: the  $F_1$  score is 0.305 at a prevalence of 0.180, 0.544 on the registered subset (0.373), and 0.452 on validation (0.292); for random Rank-1 it is 0.336, the mean average precision (mAP) is 0.385, the detection-and-identification rate (DIR) is 0.038 at a false alarm rate (FAR) of 0.1. The seeds are fixed; the spread is not reported.

Decisions: pairs above the validation threshold are accepted; open set accepts only the top gallery entry above threshold, else rejects as new ticket. Hungarian applies per block, dropping below-threshold pairs. Unregistered pairs stay in: predicted negative for  $F_1$ , ranked last; unregistered unknowns count as correct rejections while unregistered knowns count as misses, so FAR uses all 258 unknowns (102 with no registered candidate). Registration failure is absence of evidence, not evidence of a new crack. Metrics are defined in Table I caption; within-visit only. Snapshot: Mendeley DOI in [37]; score matrices and commit hash to be archived with the camera-ready version.

### V. A GEOMETRIC REFERENCE BASELINE

The test examines whether the position of the crack has an effect on the association by aligning the wall surfaces and then comparing the projected masks. It makes the assumption of a roughly planar wall and that there is no CRACKID training. This kind of procedure is typical for spatial verification; what is provided here is the protocol, the labels and the controls. The masks (UNet++ [38], [39], ResNet-34 [40]), only those that are publicly available, an area of 200 px, with a 5 px closing) are taken as given and the error is not measured.

#### A. Registration using wall texture

For  $I_a$  and  $I_b$ , we estimate a homography  $H$  (such that  $x' \sim Hx$ ) based on the surrounding wall texture, with the cracks enlarged and then omitted, so the alignment relies on surface features without accounting for the crack under examination. This independence comes from the masks and the following stage, not from each individual pair. The complete detector settings (Scale-Invariant Feature Transform (SIFT) [15] together with Contrast Limited Adaptive Histogram Equalisation (CLAHE) [41], ratio test, robust estimation) are available together with the code.

#### B. Registration cascade and validity checks

The textured render produces keypoints, whereas smooth plaster does not. For the cascade, masked SIFT is used; if the number of keypoints is too low, a fallback to unmasked SIFT is employed (in this case it is flagged and crack leakage occurs during alignment); finally, photometric alignment is carried out using the Enhanced Correlation Coefficient (ECC) [34]. The candidate solutions are tested for plausibility (they must not contain mirrored or folded quads; the scale must be within [1/16, 16]; there is no criterion based on inlier count here). We record the result of each surviving stage for each pair, and the inlier root-mean-square error (RMSE) at that stage determines the comparison tolerance. The absence of a registration for a pair means a lack of evidence, not that the cracks differ.

### C. Crack projection and geometric comparison

The Accepted Registrations project instances from  $I_a$  into  $I_b$  so they can be compared using the symmetric mean Chamfer distance [42] as in (1); this method allows small misalignments by using intersection over union (IoU) on narrow cracks. Let  $a, b$  be crack point sets and  $d(p, b) = \min_{q \in b} \|p - q\|_2$ :

$$D(a, b) = \frac{1}{2} \left[ \frac{1}{|a|} \sum_{p \in a} d(p, b) + \frac{1}{|b|} \sum_{q \in b} d(q, a) \right]. \quad (1)$$

We reject candidates that are far away (using distance in the code). Chamfer coverage (i.e., the fraction within the per-pair RMSE, with the distance clamped at [2,16] pixels and a default of 6 pixels) is the scale-free equivalent.

### D. Correspondence decisions and limits of spatial evidence

For each pair of photographs we carry out a one-to-one matching using the Hungarian method [43] at the validation threshold; pairs that fail to be registered are given a value of  $-\infty$  (Sec. IV). Alignment of an image with an empty wall acts as evidence of the absence of a target; missed detections, areas that are out of frame, and registration failures still remain unresolved. The reference targets do not show any change in damage; performance is determined by the surface and by the origin of the label.

## VI. OFF-THE-SHELF APPEARANCE BASELINES VS FULL-PHOTOGRAPH REFERENCE

### A. Setup

Ten matchers use crack-centred boxes with a common mask: SIFT/ORB with RANSAC inliers, SuperGlue/LoFTR with indoor weights, five embeddings and OSNet with cosine on L2 crops, all without CRACKID tuning (YOLOv8: layer-9 `yolov8n.pt`; sizes/checkpoints in code). Context expands the box  $1 \times$  per side; erased views inpaint via Telea after 5 px dilation (artefacts possible; sibling/neighbour handling unstated). Only OSNet/CLIP tested, no crack-only/wall-only views. REG-ONLY scores a constant on registered pairs. Appearance sees tight crops, geometry full photographs. Missing: full-photo descriptors (DINOv2, NetVLAD [24]), LightGlue [44], matcher-fed Chamfer (see repo for timings; sessions not load-controlled, no ranking claimed).

### B. Result: ranking spreads, decisions stay near the constant baseline

Ranking spreads while thresholded decisions stay near the constant predictor (Table I). Rank-1 0.352–0.654; mAP to 0.520; crop DIR to 0.429. Oracle  $F_1^*$  0.305–0.359; validated  $F_1^{\text{@}v}$  0.293–0.351 (excess  $-0.012$  to  $+0.012$  except LoFTR  $+0.046$ ; ORB/SIFT/SuperGlue accept everything at 0.305). AUROC/AUPRC, DIR–FAR curves, thresholds and wall-clustered CIs need archived matrices; leave-one-wall-out over 17 walls should replace the 6/11 split. Part of the drop is prevalence shift (0.292 to 0.180), not only appearance. With 19 identities on 11 walls, we claim no fine-grained ranking or separability from chance.

### C. Is it the crack, or the wall behind it?

For OSNet, enlarging the crop raises DIR from 0.168 to 0.404 at FAR 0.1; erasing the crack retains DIR at 0.393 (Rank-1: 0.711 vs 0.775), suggesting that location-specific context carries much of the signal. For CLIP, DIR rises from 0.143 to 0.311 with context and 0.286 after erasure, showing a similar pattern. Deltas are point estimates without CIs. The  $1 \times$  box mostly adds neighbouring cracks, possibly sibling fragments (45% merged), so the signal source is context around cracks rather than wall texture alone. Crack-only and wall-only views on additional backbones with CIs would sharpen this interpretation.  $F_1^*$  stays 0.307–0.361.

## VII. RESULTS AND DISCUSSION

### A. Main comparison and the open-set gap

For the registered cases, which make up 48% (all 5,722 positive examples), geometry is used to compare masks whereas the crops evaluate all the pairs from different pools. When registration is used alone (with REG-ONLY set to a constant value on registered cases and to  $-\infty$  in all other cases), the results are a Rank-1 score of 0.515, an mAP of 0.533, and a DIR of 0.000 at a FAR of 0.1; however, when mask comparison is added, the values improve by  $+0.437$  and  $+0.353$  respectively and the DIR reaches 0.768 (as shown in Table I). The DIR scores are 0.768 for geometry, 0.000 for REG-ONLY (with tied scores), 0.429 for SuperGlue, and 0.404 for OSNet+ctx (with a chance level of 0.038); there was no wall-clustered test and no claim of separability. The validated  $F_1$  excess is  $+0.038$  (compared to 0.582 versus 0.544) over LoFTR's  $+0.046$ , indicating that the appearance/geometry contrast is overstated. The advantage obtained through gating would disappear if cross-visit testing were carried out; further research is still needed on the gated-appearance and frame-gap baselines.

### B. Registration rate, failures, and ablations

Unregistered pairs are ranked last and predicted as negative (see Section IV). The gate is label-informative: among the registered pairs, all positives are retained, but 36.8

The comparison shows an improvement of  $+0.112$  for Oracle (with an  $F_1^*$  of 0.656) but only  $+0.038$  for the validated results (with an  $F_1^{\text{@}v}$  of 0.582). The failures depend on the wall (as described in Sec. III); for adjacent frames, all 280 queries are answered (the median gap is 1), which is why a frame baseline is required. We define sharpness as the Laplacian variance at quarter scale. A test-only logistic indicates that sharpness is better than gap (with a clustered  $z$  value of 5.11 versus  $-2.78$ ), but the  $p$  value overestimates the precision (since there are 11 clusters); the sharpness  $\geq 10$  gate is determined post hoc. The scope of the smooth-plaster failure is not quantified. The ablations for registration/ECC/exclusion were carried out on the test set; the geometry settings should be fixed on the validation set before any final comparison.

TABLE I

MAIN RESULT, TEST (11 WALLS,  $n_Q=280$  KNOWN, 258 UNKNOWN). POINT ESTIMATES ONLY; EFFECTIVE SAMPLE IS WALLS. METRICS: R@1 RANK-1 ACCURACY; MAP MEAN AVERAGE PRECISION; DIR DETECTION-AND-IDENTIFICATION RATE AT FAR 0.1 (FRACTION OF KNOWN QUERIES CORRECTLY IDENTIFIED WHEN FALSE ALARM RATE  $\leq 0.1$ );  $F_1^*$  ORACLE-THRESHOLD  $F_1$ ;  $\Delta F_1^*$  EXCESS OVER CHANCE (0.305);  $F_1^{\text{@v}}$  VALIDATION-THRESHOLD  $F_1$ ; REG. FRACTION OF TEST PAIRS THAT REGISTER. BOLD: BEST CROP PER COLUMN. GEOMETRY SCORES 48% OF VALID PAIRS AND RANKS THE REST LAST, SO DIR 0.768 VS 0.429 USE DIFFERENT CANDIDATE POOLS.

| Method                                              | Family    | R@1          | mAP          | DIR<br>@FAR = 0.1 | $F_1^*$      | $\Delta F_1^*$ | $F_1^{\text{@v}}$ | reg. |
|-----------------------------------------------------|-----------|--------------|--------------|-------------------|--------------|----------------|-------------------|------|
| <i>Chance (random ranking)</i>                      |           |              |              |                   |              |                |                   |      |
| Chance                                              | reference | 0.336        | 0.385        | 0.038             | 0.305        | 0.000          | 0.305             | 1.00 |
| <i>Reference level (same 0.48 pool as geometry)</i> |           |              |              |                   |              |                |                   |      |
| REG-ONLY                                            | geometry  | 0.515        | 0.533        | 0.000             | 0.544        | 0.000          | 0.544             | 0.48 |
| <i>Crop scope</i>                                   |           |              |              |                   |              |                |                   |      |
| ORB [16]                                            | keypoint  | 0.352        | 0.364        | 0.018             | 0.305        | 0.000          | 0.305             | 1.00 |
| SIFT [15]                                           | keypoint  | 0.432        | 0.417        | 0.036             | 0.313        | 0.008          | 0.305             | 1.00 |
| SuperGlue [17]                                      | learned   | 0.562        | 0.455        | <b>0.429</b>      | 0.353        | 0.048          | 0.305             | 1.00 |
| LoFTR [18]                                          | learned   | 0.549        | 0.440        | 0.336             | <b>0.359</b> | <b>0.054</b>   | <b>0.351</b>      | 1.00 |
| YOLOv8 backbone [23]                                | embedding | 0.561        | 0.453        | 0.157             | 0.357        | 0.052          | 0.308             | 1.00 |
| ViT-B/16 [19]                                       | embedding | <b>0.654</b> | 0.511        | 0.286             | 0.331        | 0.026          | 0.317             | 1.00 |
| DeiT-S [20]                                         | embedding | 0.621        | 0.497        | 0.250             | 0.305        | 0.000          | 0.293             | 1.00 |
| CLIP ViT-B/32 [21]                                  | embedding | 0.604        | 0.471        | 0.143             | 0.307        | 0.002          | 0.306             | 1.00 |
| DINOv2 [22]                                         | embedding | 0.607        | 0.484        | 0.307             | 0.305        | 0.000          | 0.297             | 1.00 |
| OSNet [3]                                           | re-ID     | <b>0.654</b> | <b>0.520</b> | 0.168             | 0.309        | 0.004          | 0.309             | 1.00 |
| <i>Full-photograph scope</i>                        |           |              |              |                   |              |                |                   |      |
| REG+CHAMFER (reference)                             | geometry  | 0.952        | 0.886        | 0.768             | 0.656        | +0.112         | 0.582             | 0.48 |

### C. Synthetic revisit probe

From five walls, using one seed, ten source photographs show a small warp either under (A) one-tip growth or (B) blur/noise/contrast with the structure remaining unchanged, and then they are matched with the actual images (p90 tilt 75°; procedure as in the code). HYBRID-SKEL ( $\tau=8.0$  px) differs from REG+CHAMFER (see Table I). In the case of ranking/open set, OSNet+ctx outperforms HYBRID-SKEL (R@1 0.826/0.671 versus 0.275/0.274; DIR 0.739/0.663 versus 0.343/0.342;  $F_1^{\text{@v}}$  is similar at 0.601/0.594 versus 0.569/0.574); assignment- $F_1^a$  shows a narrow reversal (0.646/0.640 versus 0.631/0.609; there is no pure-homography cell). With respect to RQ3, this single-seed probe provides no evidence of real multi-visit robustness, which has not been tested (Sec. VIII).

## VIII. LIMITATIONS AND FUTURE WORK

To our knowledge, this study is the first measurable single-visit crack re-ID test with a geometric reference. Threats: (i) 19 identities on 11 walls, queries not independent, CIs wide; (ii) labels and winner share homography, possible circularity; (iii) different pools with label-informative gate; (iv) single-threshold  $F_1$  at 0.180 near-degenerate with validation/test mismatch; (v) singleton unknowns, unstratified variation, wall given but location, Building Information Modelling (BIM), physical markers, and Structure from Motion (SfM) are unevaluated.

- 1) **More walls and visits.** Add walls, surfaces, annotators (CIs, paired tests); add a second session for cross-visit;
- 2) **Labels without homography.** Small set without overlays ( $\kappa$ , ARI), re-run all methods, merged-vs-unmerged, plus matrices for CIs and AUROC/AUPRC.

- 3) **Registration coverage.** The reference registers 97 out of 301 of the same-wall test photograph pairs (32% of the photograph pairs; 48% of the 15,328 valid query/gallery pairs within those pairs) but does not work on smooth-plaster walls, since the median sharpness in such cases is low (see Section III). Coverage could improve with surface-adaptive, photometric, or depth-based registration, which would also separate geometry and labelling; for the FAR measure to assess novelty, matched unknowns from the holdout set are required.
- 4) **Tickets.** No ticket-level duplicate rate is measured, and the problem that is causing concern is stated without referring to any ticket data. The definitions of duplicate rate and missed-new rate should be established, and the segmentation and assignment processes should be tested in a closed loop.

Together, these turn the within-visit reference into a multi-visit ticket component.

## IX. CONCLUSION

We present CRACKID, which consists of 140 photographs from 17 walls, along with a protocol for retrieval, deciding whether defects are the same, and performing open-set identification with rejection.

The key results are positive: with regard to (RQ1), ten of the zero-shot crop matchers perform better than random, thus setting a viable baseline for retrieval, while thresholded decisions stay near the level of the best constant baseline and therefore call for a combined assessment of ranking, decisions, and rejection; as for (RQ2), local wall context makes a significant contribution to appearance-based association, and the geometric reference shows how surface alignment and

projected crack overlap can assist open-set identification on the 48% alignable subset; regarding (RQ3), the small synthetic revisit probe makes clear the distinction between within-visit association and multi-visit robustness.

The fact that good retrieval performance alone is not enough to ensure reliable crack association shows the need for both local visual context and geometric alignment to perform re-identification. Because CRACKID makes judgments not only on retrieval but also on matching and on rejecting unknown defects, CRACKID forms a basis for the development of more reliable visual association methods to support defect tracking in ticket management systems.

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